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ANALYSIS **Open Access**

Dexmedetomidine and dezocine targets (NR1H3 and ADRB1) are potential prognostic biomarkers for breast cancer

Shaochun Zhang[1*†], Fang Yuan[2†] and Jing Ke[3]

†Shaochun Zhang and Fang Yuan have contributed equally to this work.

*Correspondence:
Shaochun Zhang
sczhangdoctor@126.com
[1]Department of Orthopedics, Ezhou Central Hospital, Ezhou, Hubei, China
[2]Department of Anesthesiology, Ezhou Central Hospital, Ezhou, Hubei, China
[3]Department of Endocrinology, Ezhou Central Hospital, Ezhou, Hubei, China

Abstract
Breast cancer (BRCA) is a prevalent malignant tumor among women, and the use of anesthetic drugs during surgical resection may influence tumor biology and patient prognosis. This study aimed to identify prognostic biomarkers associated with dexmedetomidine and dezocine (DD) in BRCA patients. Through Mendelian Randomization analysis, we screened four DD targets that had a causal relationship with BRCA. Subsequently, utilizing TCGA-BRCA data, univariate and Lasso Cox analyses revealed two significant prognostic biomarkers (NR1H3 and ADRB1) associated with BRCA patient prognosis, leading to the successful construction and validation of a prognostic risk model. Kaplan-Meier survival curves indicated that patients with higher NR1H3 and ADRB1 expression had longer overall survival (OS). Immunoinfiltration analysis showed that high-risk group patients exhibited increased infiltration levels of CD56 bright natural killer cells, CD56 dim natural killer cells, eosinophils, and plasmacytoid dendritic cells. Conversely, activated B cells and immature B cells demonstrated greater infiltration in the low-risk group. Correlation analysis revealed significant associations between prognostic biomarkers and various immune cells, including CD56 bright natural killer cells, CD56 dim natural killer cells, and activated CD8 T cells. NR1H3 was highly positively correlated with immune checkpoints such as TIGIT, PDCD1, CD274, CTLA4, LAG3, and HAVCR2 ($|cor|\geq 0.3$, $P<0.05$). Immunohistochemistry images and Transcripts Per Million values from the Human Protein Atlas database confirmed that NR1H3 and ADRB1 expression levels were significantly higher in normal tissues compared to BRCA tissues. Our results suggest that target genes of anesthetic drugs (NR1H3, ADRB1) may be involved in BRCA prognosis.

Keywords BRCA, Dexmedetomidine and dezocine, MR, Prognostic risk model, Immunoinfiltration, Biomarkers

1 Introduction

Breast cancer (BRCA) is one of the most common malignancies in women worldwide and the fifth leading cause of cancer deaths globally [1]. China has the highest number of BRCA cases in the world, accounting for approximately 20% of global BRCA cases [2].

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The incidence and mortality rates of BRCA have been increasing year by year [3], posing a serious threat to women's lives and health [4].

Surgical excision remains the cornerstone of treatment for BRCA. However, despite complete tumor resection, approximately one-third of patients experience recurrence and metastasis, which directly account for mortality in this population [5, 6]. Clinical practice has demonstrated that anesthetic drugs used during surgery in BRCA patients may impair the body's immunity and affect the biological behavior of the tumor and patient prognosis [7].

Previous studies have shown that inhalation anesthetics such as isoflurane and sevoflurane can inhibit natural killer cell and t-lymphocyte-mediated immunity, leading to an increase in BRCA metastasis [8–10]. Dexmedetomidine could promote the proliferation, migration and invasion of BRCA cells through the activation of α2B-adrenoceptor /ERK signaling [11]. In addition, dexmedetomidine down-regulated the expression of miR-199a in BRCA cells and enhances the down-regulation of miR-199a to promote the progression of BRCA [12]. Animal experiments have pointed towards the potential role of dexmedetomidine in promoting cancer recurrence and metastasis when used perioperatively especially after breast surgeries [13]– [14]. Dexmedetomidine is an α2-adrenergic agonist, which exists in the pre-synaptic membranes of the central nervous system, and through stimulation of the α2-adrenergic receptors, it can control the production of catecholamines in the precordium of nerve cells, which can have a better anesthetic effect on patients, shorten the recovery time of patients, and improve the safety of anesthesia for patients [15]. Dezocine inhibits the proliferation, migration, and invasion of triple-negative breast cancer (TNBC) cells, induces apoptosis of TNBC cells, and possesses some antitumor activity [16]. Dezocine is an atropine antagonist, which can greatly reduce the symptoms of postoperative severe pain, and has been widely used to alleviate the pain of postoperative therapy [17]. These two drugs are often used in BRCA surgical anesthesia. Clinical trials have confirmed that dexmetopmine combined with dezocine can provide significant clinical efficacy for patients undergoing general anesthesia surgery, and contribute to the early recovery of immune function in patients after mastectomy. Liu [18] et al. found that dexmedetomidine combined with dezocine could effectively maintain the hemodynamic stability of patients undergoing BRCA surgery, alleviate inflammatory stress response and immunosuppression, and had no significant impact on postoperative anesthesia recovery, showing good anesthetic effect and safety. Zhao et al. [19] and Xie et al. [20]. found that dezocine combined with dexmedetomidine could reduce the degree of agitation during the recovery period of BRCA patients undergoing radical mastectomy, and alleviate immunosuppression and stress response. These studies indicate that dexmedetomidine and dezocine (DD) not only participate in the proliferation and metastasis of BRCA cells, but also participate in the immunosuppressive response, and play an important role in the progression of BRCA. However, at present, the mechanism of action between these two narcotic drugs and BRCA and whether they affect the prognosis of BRCA patients have not been fully clarified.

Based on this, the DD targets data were obtained from PubChem platform (https://pubchem.ncbi.nlm.nih.gov/) and the PharmMapper platform (http://www.lilab-ecust.cn/pharmmapper/), SNP data from BRCA patients and anesthetic targets and transcriptome and clinical database were obtained from the Genome-Wide Association Studies (GWAS) database, from Gene Expression Omnibus (GEO) and the Cancer Genome

Atlas (TCGA) database. Systematically analyzed the causal relationship between DD targets and BRCA, and identified BRCA prognostic biomarkers related to BRCA patients' prognosis.

2 Methods

2.1 Data availability

Gene expression profiles of human BRCA were obtained from the GEO (https://www.ncbi.nlm.nih.gov/geo/). The GSE42568 dataset has 104 breast tissue from patients and 17 normal breast tissue. We obtained the mRNA expression database TCGA-BRCA and the corresponding clinical data of BRCA patients from the TCGA (https://portal.gdc.cancer.gov/repository) database.

2.2 Screening of target genes of DD

The 2D and 3D structures and targets of DD were obtained from PubChem platform (https://pubchem.ncbi.nlm.nih.gov/) and the PharmMapper platform (http://www.lilab-ecust.cn/pharmmapper/) .

2.3 Identify differentially expressed genes (DEGs) between the BRCA and the normal group

The GSE42568 dataset was utilized to identify DEGs using R (4.4.1) "Limma" package between the BRCA and the normal group. The R "VennDiagram" package was utilized to identify the intersection of DEGs and targets, resulting in the identification of candidate targets associated with BRCA.

2.4 Mendelian randomization (MR) analysis

2.4.1 Data acquisition of exposure factors and outcomes

(1) The candidate targets-related eQTL data was downloaded from the website (https://www.gtexportal.org/home/) and converted it into MR format data. The eQTL data was extracted for the candidate targets from BRCA tissue and recorded as the candidate targets-SNP

(2) BRCA SNP datasets were obtained from the GWAS database: ebi-a-GCST90018799, and read the BRCA-SNP using the "format_data" function in the R (4.4.1) "TwoSampleMR" package.

2.5 Selection of instrumental variables (IVs)

To obtain effective IVs for subsequent analysis, the extract_instruments function in R (4.4.1) " TwoSampleMR" package is employed to read exposure factors and outcome events and screen the IVs. The R (4.4.1) "ieugwasr" function ld_clump was used to filter candidate gen-SNPs, remove SNPs with linkage imbalance (clump = TRUE, $r2 = 0.001$, $kb = 10000$), and remove weak tool variables (F-statistics < 10). SNPs strongly correlated with BRCA were obtained and recorded as instrumental variables.

2.6 MR analysis

To identify characteristic genes that can serve as intervention targets for BRCA, SNPs were defined as IVs, candidate targets as exposure factors, The MR Analysis was performed using the TwoSampleMR package of R (4.4.1) software. Five algorithms of MR were utilized to conduct univariate and multifactorial MR Analysis (MR Egger, Weighted median, Inverse variance weighted (IVW), Simple mode, Weighted mode) for each candidate exposure factor and outcome. IVW was employed as the main analysis method in this study. Horizontal pleiotropy analysis was carried out using MR-Egger intercept, and Cochran's Q statistic along with corresponding P-value were used to assess heterogeneity. A P-value > 0.05 indicated that heterogeneity was not significant. Finally, a leave-one-snp out analysis was performed to evaluate result reliability and potential SNP impact.

2.7 Selection of prognostic biomarkers and prognostic risk model construction

To identify targets associated with overall survival (OS) for BRCA patients, based on TCGA-BRCA, we used univariate Cox regression analysis to screen targets related to OS for BRCA patients. Then, we utilized Lasso Cox regression to identify biomarkers associated with the prognosis of BRCA patients based on performed penalty parameter adjustment through 10-fold cross-validation via 'Glmnet' R (4.4.1) and construct prognosis models.

$$\text{Risk score} = \sum_{n=1}^{\infty}(\text{coef}_1 * \text{exp}_1 + \text{coef}_2 * \text{exp}_2)$$

The coef indicated the regression coefficients, exp indicated the expression level of biomarkers in BRCA patients,1,2 indicated the symbol of biomarkers.

BRCA patients were divided into high-low risk groups based on a median risk score, and evaluated the stability of the prognostic risk model by GSE42568 datasets. The Kaplan-Meier (KM) survival curve was drawn by R"survival" package to predict patient OS. Receiver operating characteristic (ROC) curves were plotted to assess the accuracy of the prognostic model. To further analyze the effects of prognostic biomarkers on BRCA patients OS, Interactive gene expression profiling (GEPIA) (http://gepia.cancer-pku.cn) website was selected to plot the OS curve.

2.8 Analysis of independent prognostic value and nomogram construction

In order to determine that risk score can be used as independent factors in predicting BRCA, based on TCGA-BRCA dataset, all clinicopathological features [21] and risk score were incorporated into the prognostic model for univariate and multivariate Cox analysis to determine the independent predictive value of risk score. Then, the R "rms" package was used to construct a nomogram, and the nomogram was used to score the risk score and clinical characteristics respectively. Each factor corresponds to a score, and the total score of each factor was added to correspond to the total score. Then the survival probability of 1, 3 and 5 years was predicted according to the total score. To verify the accuracy of the nomogram, calibration curves and decision curves of the 1-year, 3-year and 5-year nomogram were drawn based on the prognostic model to verify the validity of the nomogram. The calibration curves were distributed on both sides of the diagonal line, and the closer the slope to the diagonal line, the more accurate the prediction.

2.9 Evaluation of the tumour immune microenvironment using the SsGSEA algorithm

The ssGSEA algorithm quantified the infiltration of 28 immune cell types using the normalized enrichment score (NES). An independent sample t-test compared infiltration between the high-low risk groups, with $P < 0.05$ deemed statistically significant.

2.10 Correlation analysis between prognostic biomarkers and immune checkpoint

Since dexmedetomidine combined with dezocine can reduce immunosuppressive responses, it is necessary to explore the correlation between prognostic biomarkers and immune checkpoints(SIGLEC15, PDCD1LG2 (PD-L2), TIGIT, PDCD1 (PD-1), CD274 (PD-L1), CTLA4, LAG3 and HAVCR2 (TIM3)) [22]. In order to explore the relationship between the high-low risk group and immune checkpoint, based on the TCGA-BRCA dataset, immune checkpoints gene with $P < 0.05$ was selected for subsequent analysis by comparing the differential expression of immune checkpoints gene between the high-low risk group, and "ggplot2" was used to draw a box plot for visualization. In order to explore the correlation between prognostic targets and differential immune checkpoints genes, Spearman correlation analysis was performed on the prognostic targets and differential immune checkpoints genes using the R package "psych" based on the samples of the high-low risk groups in the TCGA-BRCA dataset. The heat map was drawn for visualization.

2.11 Prognostic biomarkers expression level validation

From human protein mapping (HPA, https://www.proteinatlas.org/) predict prognosis biomarkers expression levels.

2.12 Statistical analysis

Bioinformatics analysis was performed using R software (4.4.1). The Wilcoxon test analyzed differences between different groups. Unless otherwise stated, $P < 0.05$ or $P < 0.05$ indicated a significant difference.

3 Result

3.1 Identification of DD targets associated with BRCA

In order to obtain DD targets associated with BRCA, we analyzed the transcriptome data of GSE42568 and screened a total of 3383 DEGs in the BRCA patients breast groups and the control groups, of which 1780 were up-regulated and 1603 were down-regulated (Fig. 1A, B); we obtained DD targets from PubChem platform and the PharmMapper platform (Supplementary Table 1). Intersecting the DEGs with targets, we obtained a total of 28 DD candidate targets (Supplementary Table 2)associated with BRCA (Fig. 1C).

3.2 Identification of targets causal associated with BRCA

Using MR Egger, Weighted median, Inverse variance weighted (IVW), Simple mode and Weighted mode, the causal relationship between candidate targets and BRCA was analyzed. We found that 4 targets (NR1H3, ADRB1, TLR4, and EPHX1) were significantly association with BRCA based on the IVW approach (all p value < 0.05) (Fig. 2A, B). Evidently, from Fig. 2C, The SNP of NR1H3, ADRB1, TLR4 reduced the risk of BRCA, the SNP of EPHX1 increased the risk of BRCA. The IVW approach revealed

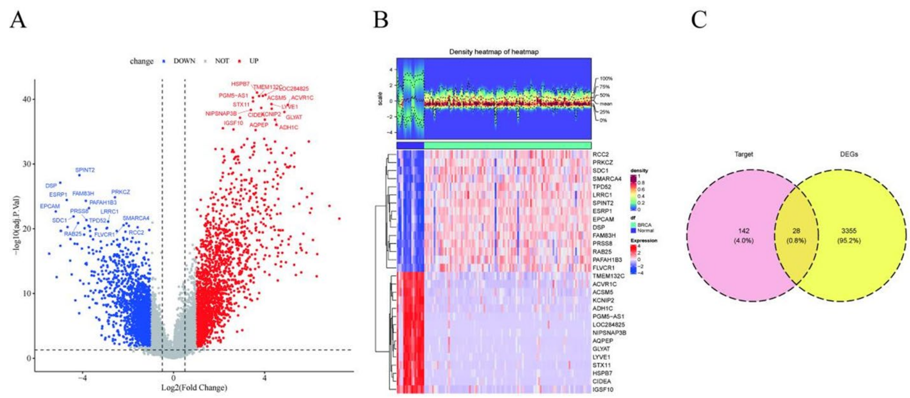

Fig. 1 Identification targets of DD associated with BRCA. **A** Volcano plot displaying the DEGs between BRCA and control groups. Red dots denote upregulated genes ($log_2FC > 1$, FDR < 0.05), blue indicates downregulated genes ($log_2FC<-1$, FDR < 0.05), with gray representing non-significant genes (FDR ≥ 0.05). Dashed lines mark thresholds at ± 1 log_2FC. **B** Hierarchical clustering heatmap of the top 20 DEGs across BRCA ($n = 104$) and normal ($n = 17$) specimens. The upper dendrogram reflects Euclidean distance-based sample clustering, while the lower color matrix visualizes Z-score normalized expression patterns (red: high expression; blue: low expression). **C** Venn diagram demonstrating intersection between DEGs and targets of DD. The overlapping region ($n = 28$) represents candidate targets subjected to subsequent MR analysis

no heterogeneity among the SNPs significantly associated with BRCA ($P > 0.05$) (Supplementary Table 3). The funnel plot's impact on causation was roughly symmetrical (Fig. 2D). The MR-Egger-regression showed no directional pleiotropy (Supplementary Table 4). The leave-one-out analysis showed that a single SNP did not drive the MR analysis results (Fig. 2E).

3.3 Prognostic model construction and validation

Two prognostic biomarkers (NR1H3, ADRB1) associated with OS for BRCA patients were identified using univariate(Fig. 3A)and Lasso Cox regression analyses (coef: NR1H3=-0.3703934; ADRB1=-0.4123436)(Fig. 3B)base on TCGA-BRCA data. We calculated the risk score and constructed the prognostic risk model based on two prognostic biomarkers, and the distribution graphs of the risk scores showed that the shortening of the survival time was accompanied by an increase in the risk scores (Fig. 3C, D); Fig. 3E showed the expression levels of prognostic biomarkers in the high-low risk groups in TCGA-BRCA. The survival curves showed (Fig. 3F) that the OS of low-risk patients were all better than that of high-risk patients ($P < 0.05$, log-rank test); ROC curves (Fig. 3G) revealed the AUC value of the 5-year OS of the model was 0.626, indicating that the prognostic risk model has discriminatory ability. Patients were divided into high and low-risk groups according to the risk score in GSE42568. Figure 4A, B,D showed that the OS of the patients in the high-risk group was lower than that of the patients in the low-risk group. Figure 4C showed the expression levels of prognostic biomarkers in the high-low risk group in GSE42568. The AUC value of the ROC curves (Fig. 4E) were 0.607, which indicated that the prognostic model as having limited discriminatory ability. KM survival analysis (Fig. 4F, G) based on GEPIA database showed that patients with high expression of NR1H3 and ADRB1 had longer OS ($P = 0.034$ and $P = 0.036$), indicating that NR1H3 and ADRB1 were protective genes of BRCA.

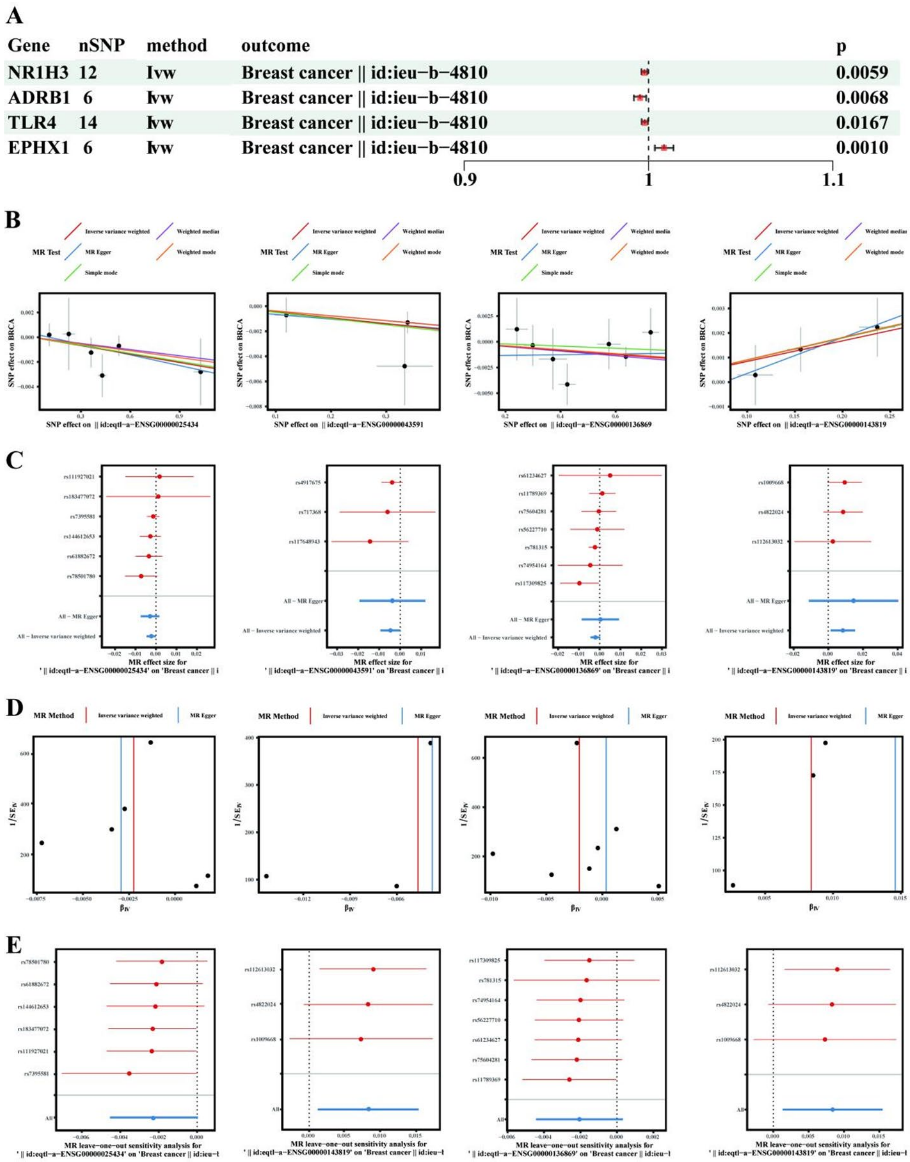

Fig. 2 The MR analysis results between targets of dexmedetomidine and dezocine and BRCA. Forest plot of IVW MR analysis. **B**. Scatter plot of genetic association effect sizes for instrument-target (x-axis) versus target-BRCA (y-axis) relationships. The weighted linear regression line represents causal effect magnitude. **C**. Forest Plot. The name of each SNP is displayed on the left. Each SNP corresponds to one row in the plot. The X-axis represents the interval of the effect size. The black dots in each row indicate the magnitude of the effect size for the respective SNP. Black dots located to the right of the no-effect line (the vertical black dashed line) suggest that exposure may increase the likelihood of the outcome, whereas those on the left side indicate a potential reduction in the occurrence of the outcome due to exposure. The length of the line segment represents the 95% CI. If the line segment crosses the zero-dashed line, it implies that the result is not statistically significant. Conversely, if the entire segment lies entirely to the left or right of the zero-dashed line, it indicates statistical significance. **D**. Funnel plot evaluating pleiotropy through effect estimate symmetry. Standard error (SE) of SNP-specific effects plotted against causal estimates. **E**. Leave-one-out sensitivity forest plot assessing robustness of causal estimates. Sequential exclusion of individual SNPs demonstrates consistent directional effects, with all leave-one-out estimates remaining statistically significant. Dashed vertical line indicates primary IVW estimate

3.4 Independent prognostic analysis

Univariate Cox regression analysis based on TCGA-BRCA data (Fig. 5A)showed that age, risk score, and stage were significantly associated with the prognosis of BRCA. Multifactorial Cox regression analysis (Fig. 5B)showed that risk score remained an independent

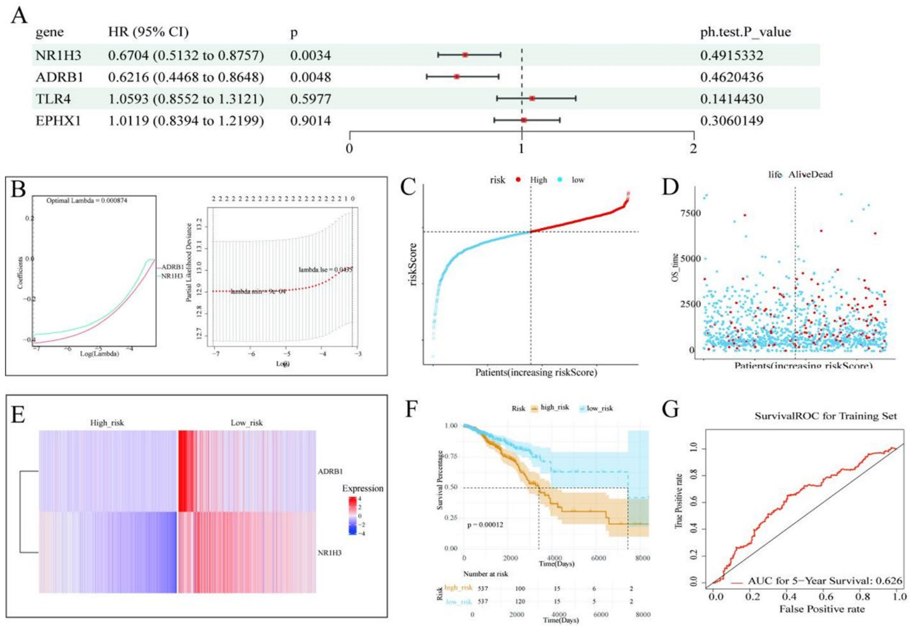

Fig. 3 Development and validation of the prognostic risk model. **A** Univariate Cox proportional hazards regression screening of prognosis-associated genes in the TCGA-BRCA cohort. Genes meeting significance threshold (log-rank $P < 0.01$, FDR < 0.05) are highlighted in red (hazard ratio [HR] > 1). **B** ASSO Cox regression with 10-fold cross-validation for feature selection. The vertical dashed line indicates the optimal λ value (λ = 0.0435) minimizing partial likelihood deviance, yielding 2 prognostic genes. **C** Risk score distribution stratified by median cutoff. Kernel density curves demonstrate significant divergence between high- and low-risk groups. **D** Survival status scatter plot across risk quartiles. Circle size corresponds to overall survival (OS) duration (years), with red/blue indicating deceased/censored events. **E** Hierarchically clustered heatmap of prognostic biomarkers expression (Z-score normalized). **F** K-M survival curves comparing high-vs.low-risk groups. **G** Time-dependent ROC analysis evaluating predictive accuracy (AUC: 5-year = 0.626). Dashed diagonal line indicates random prediction reference

prognostic factor after adjusting for other clinicopathologic factors. Nomograph(Fig. 5C) was constructed based on risk scores and clinical indicators. The DCA curves(Fig. 5D) showed that the column charts could produce more net gains, which might contribute to better clinical management; The calibration curves(Fig. 5E)fitted well with the ideal model, were distributed on both sides of the diagonal line, and the slopes were close to the diagonal line, which indicated that the predictions were accurate.

3.5 Analysis of immune infiltration

Immune microenvironment analysis based on TCGA-BRCA data (Fig. 6A) revealed that except for central memory CD4 T cell, Memory B cell, and Type 2 T helper cell, other cells were differentially expressed in high and low risk groups ($P < 0.05$). Figure 6B showed the correlation between prognostic biomarkers, risk scores and immune cells, indicating significant associations between prognostic biomarkers and CD56 bright natural killer cells, CD56dim natural killer cells, activated CD8 T cells and other immune cells. To further assess the relationship between prognostic biomarkers and immunotherapy, we compared the differential expression of eight immune checkpoints in the high-risk and low-risk groups, and found NR1H3 was highly positively correlated with immune checkpoints (TIGIT, PDCD1, CD274, CTLA4, LAG3, and HAVCR2) ($|cor| \geq 0.3$, $P < 0.05$), while ADRB1 exhibited litter lower correlations with these checkpoints ($|cor| < 0.3$, $P < 0.05$) (Fig. 6C-D).

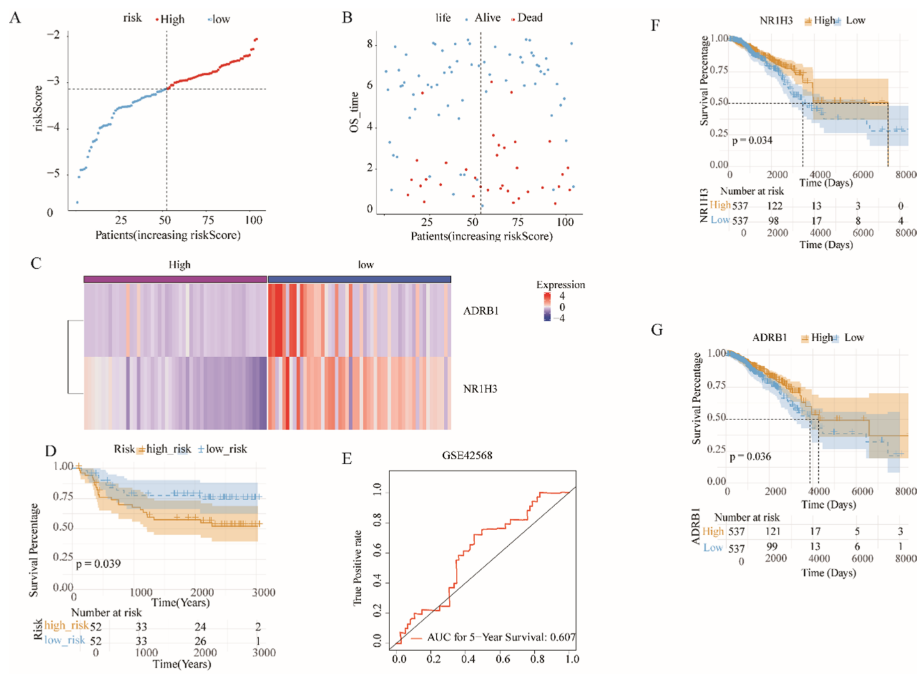

Fig. 4 External validation of the prognostic model in the GSE42568 cohort ($n = 121$). **A** Risk score distribution stratified by the predefined training set cutoff. **B** Survival status mosaic plot demonstrating risk-dependent mortality patterns. **C** Consensus clustering heatmap of prognostic biomarkers expression. **D**.K-M survival analysis calibrated by inverse probability weighting. High-risk patients show significantly reduced median OS. **E**. Time-dependent ROC analysis demonstrating temporal predictive accuracy (AUC: 5-year = 0.607). **F** K-M survival analysis of NRIH3 by GEPIA. **G**.K-M survival analysis of ADRB1 by GEPIA

3.6 Expression verification of prognostic biomarkers

IHC images and TPM values (Fig. 7A, B) from the HPA database confirmed that NR1H3 and ADRB1 expression levels were significantly higher in normal tissues compared to BRCA tissues, consistent with findings from the GSE42568 (Fig. 7C) and TCGA-BRCA (Fig. 7D) datasets used in our study.

4 Discussion

The relationship between anesthesia and tumor recurrence has been a highly debated issue in the field of surgical oncology. Currently, there is no sufficient evidence to demonstrate the ability to correlate anesthetic drugs with recurrence rates or long-term prognosis in patients undergoing BRCA surgery. The TCGA database is a public, large-scale cancer genome database containing multi-dimensional cancer-related data such as whole genome sequencing data, proteomic data, methylation data, mRNA data and miRNA data from 33 major cancer types. These data come from multiple institutions and top researchers with high reliability and authority. Therefore, this study innovatively used TCGA database to identify the dexmedetomidine and dezocine biomarkers association with BRCA, and found that four anesthetic biomarkers (NR1H3, ADRB1, TLR4, and EPHX1) were significantly correlated with BRCA, and that these four targets may play important roles in BRCA development or participate in regulating the proliferation and metastasis of BRCA cells.

To further investigate the relationship between the four anesthetic drug targets and the prognosis of BRCA patients, this study used one-way Cox regression analysis to

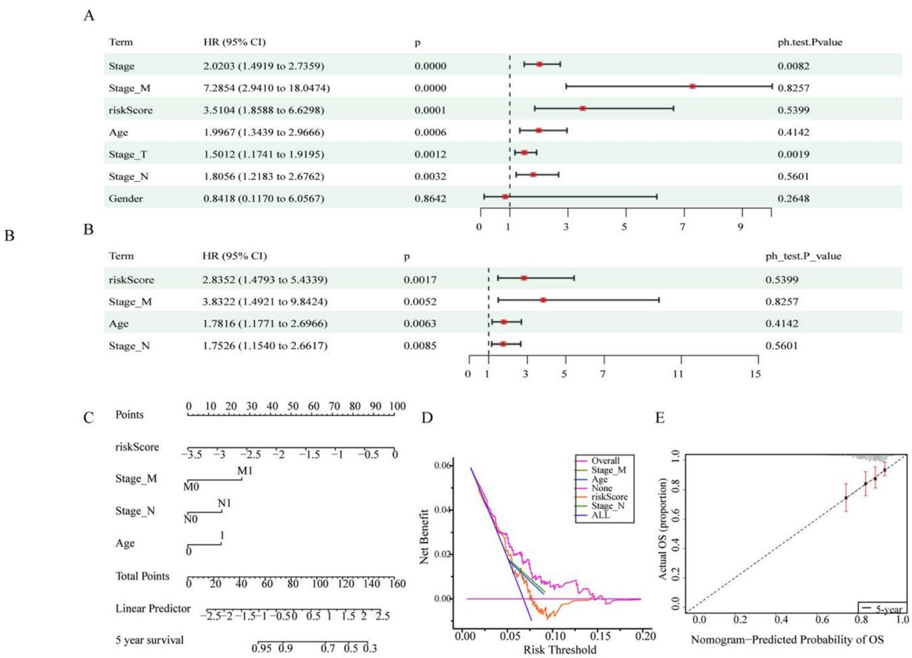

Fig. 5 Independent prognostic value and clinical utility of the risk model. **A** Univariate Cox proportional hazards regression evaluating associations between clinicopathological variables (age, TNM stage, histological grade) and overall survival (OS) in the TCGA-BRCA cohort. The risk score showed the highest independent prognostic hazard ratio in univariate Cox regression (HR = 1.92, 95% CI: 1.45–2.54, *P* < 0.001). **B** Multivariate Cox model adjusted for confounders confirmed risk score as an independent prognostic factor. Proportional hazards assumption validated via Schoenfeld residuals. **C** Nomogram integrating risk score with TNM stage for 5-year survival prediction. **D** DCA quantifying clinical net benefit across threshold probabilities. **E** 5-year calibration plot with bootstrapping. Observed vs. predicted survival probabilities showed strong agreement. The dashed 45 line represents ideal calibration

correlate the four genes with the survival and prognosis of BRCA patients, and found that NR1H3 and ADRB1 were significantly correlated with the prognosis of BRCA patients. TLR4 has been reported to play an important regulatory role in BRCA growth, invasion, and metastasis [23]. In this study, no significant correlation was found between TLR4 and prognosis of BRCA patients. While the correlation between EPHX1 polymorphisms and BRCA risk is controversial [24, 25], and has not yet been conclusively determined, the results of the analysis in this study found no significant correlation between EPHX1 expression and BRCA prognosis. Therefore, in this study, we developed a prognostic model based on two biomarkers (NR1H3 and ADRB1) significantly associated with BRCA patient outcomes. While the risk score derived from these biomarkers demonstrated modest discriminatory ability in external datasets, our findings suggest that NR1H3 and ADRB1 may serve as potential biomarkers for predicting BRCA prognosis; meanwhile, there is also a correlation between NR1H3 and ADRB1 and immune cells NR1H3 was significantly correlated with immune checkpoints and ADRB1 was less correlated with immune checkpoints. This study suggested that target genes of anesthetic drugs (NR1H3, ADRB1) may be involved in BRCA prognosis.

NR1H3 is an important member of the nuclear receptor family and a key regulator of macrophage function. Previous studies have confirmed that NR1H3 is involved in the development and progression of BRCA and affects the prognosis of BRCA patients. Low NR1H3 expression was associated with poorer survival, and BRCA patients with low NR1H3 expression had shorter overall survival and poorer prognosis [26]. This is

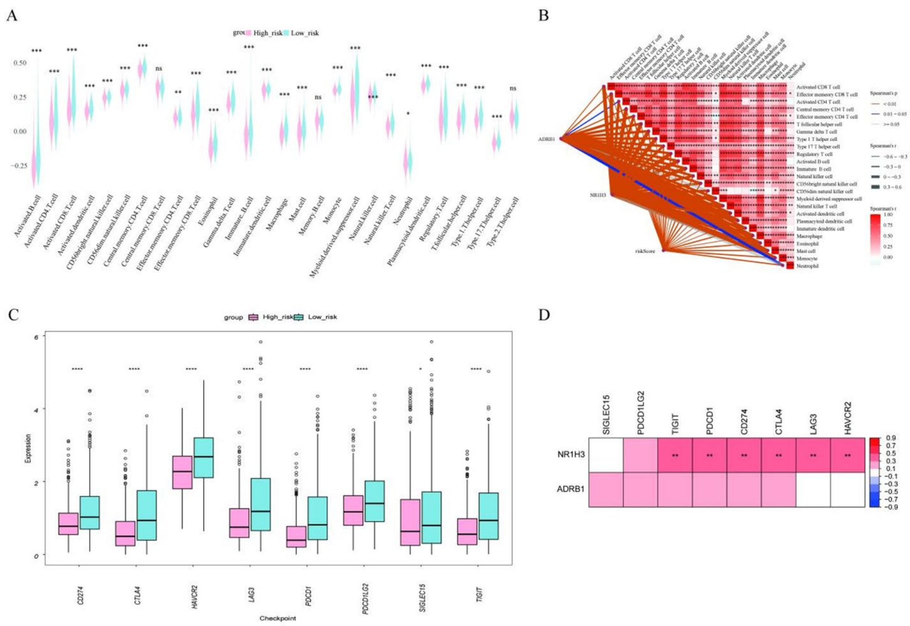

Fig. 6 Tumor immune microenvironment characteristics across risk strata. **A** Differential immune cell infiltration quantified by ssGSEA. **B** Spearman correlation network between risk score, prognostic genes, and immunocytes (|cor|>0.3, $P < 0.05$). Edge thickness reflects correlation strength, with red/blue indicating positive/negative associations. **C** Differential expression of 12 immune checkpoint genes between risk groups. **D** Correlation heatmapbetween prognostic biomarkers (NR1H3/ADRB1) and immune checkpoints. Significant correlations (Pearson $r > 0.4$, $^*P < 0.05$) marked with asterisks

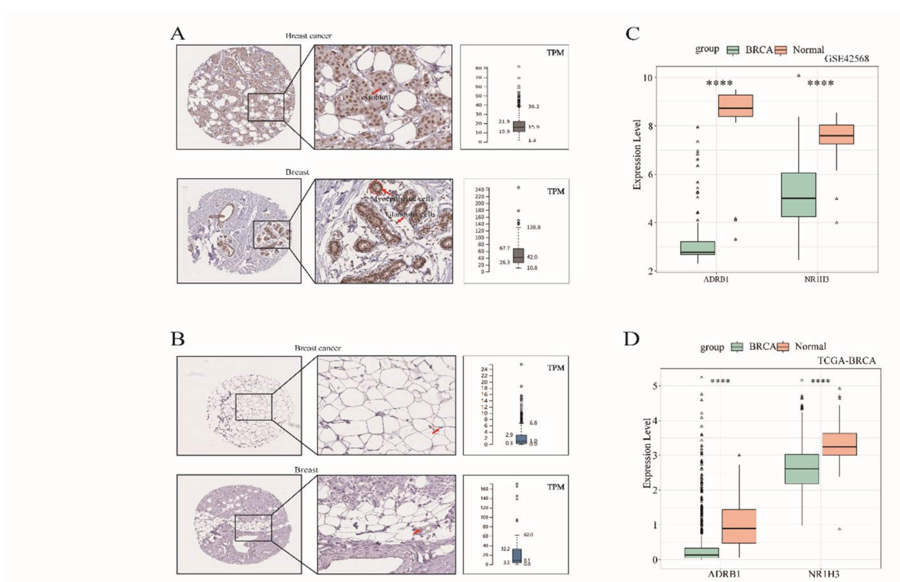

Fig. 7 Validation of prognostic biomarkers. **A**-**B**. IHC validation from HPA. **A** NR1H3 protein expression in BRCA tissue vs. normal breast. **B** ADRB1 protein expression in BRCA tissue vs. normal breast. Boxplots display median-centered TPM values. **C**-**D**. Transcriptomic concordance across platforms. **C** ADRB1 and NR1H3 expression in GSE42568. **D** ADRB1 and NR1H3 expression in CGA-BRCA

consistent with the results analyzed in this study. In BRCA, NR1H3 may be a tumor suppressor gene [27] that is associated with immune infiltration [28]. The same analytical results were obtained in this study. And NR1H3 is the target of dezocine action. Dezocine can induce apoptosis in triple-negative BRCA cells, which suggests that dezocine may affect the prognosis of patients by inhibiting cell proliferation of BRCA cells through NR1H3.

ADRB1 is a member of the G protein-coupled receptor superfamily [29, 30]. It has been reported that ADRB1 may inhibit tumor growth by causing an increase in immune cell infiltration in TME [31]. ADRB1 expression level was positively correlated with immune cell infiltration in BRCA, especially T cells [32]. This is consistent with the results of the present study, and similarly, we found that the survival of patients with low expression of ADRB1 was poorer, and ADRB1 plays a potentially protective function in BRCA by regulating the tumor immune microenvironment [33]. In addition, ADRB1 is a co-interacting agent of dexmedetomidine and dezocine, suggesting that ADRB1 has an important function in BRCA development.

It is important to recognise that, despite the important findings of our study, there are several important limitations. (1) Although MR is a powerful tool for studying causal relationships between different types of phenotypes. However, our MR analyses were based on association data between phenotypes and SNPs derived from GWAS data, and were limited by the sample size of BRCA in current GWAS data. We ensured the independence of instrumental variables through multiple validation approaches. It should be noted that MR analyses based on GWAS summary statistics inherently cannot fully exclude all potential horizontal pleiotropic pathways. Future studies should employ multi-omics approaches, such as single-cell sequencing and proteomics, to validate the biological independence of these targets at a more refined molecular level. (2) Although we have identified NR1H3 and ADRB1 as potential prognostic biomarkers with significant expression level differences confirmed through the HPA database, these findings are still only preliminary predictions based on data analysis. Further biological experiments are needed to explore and validate the specific mechanisms by which these prognostic markers influence BRCA. (3) The identified and prognostic biomarkers in this study were determined based on publicly available data sets. The samples in these public databases have limitations in terms of ethnic diversity. For instance, the transcriptome datasets sourced from TCGA mainly originate from countries with a Caucasian population. Therefore, the transcriptome datasets from different geographical regions are required to validate the the biological functions of prognostic biomarkers, and a large number of clinical experiments are also needed to verify the reliability of prognostic biomarkers.

5 Conclusion

Our results indicated that biomarkers (NR1H3 and ADRB1) can serve as potential biomarkers of BRCA patients' prognosis.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1007/s12672-025-03694-7.

Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

Supplementary Material 4.

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Author contributions

Shaochun Zhang carried out the studies, participated in collecting data, and drafted the manuscript. Fang Yuan performed the statistical analysis and participated in its design. Jing Ke participated in acquisition, analysis, or interpretation of data and draft the manuscript. All authors read and approved the final manuscript.

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Data availability

The datasets analysed during the current research are all available in the GEOhttps://www.ncbi.nlm.nih.gov/geo/ databases and TCGA https://portal.gdc.cancer.gov/repository database.

Declarations

Ethics approval and consent to participate

Not applicable.

Competing interests

The authors declare no competing interests.

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